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Question

Initially observed in 2017, the interstellar object 'Oumuamua is the first object of its kind to be seen in our solar system. Researchers have been puzzled because its acceleration cannot be entirely explained by the gravitational pull of nearby bodies: there must be a nongravitational influence on its velocity and trajectory. Some previously suggested explanations for this nongravitational acceleration involve mechanisms that are unlikely or unrealistic, such as geometric effects from 'Oumuamua being potentially composed of several spatially separated bodies. Now, Jennifer Bergner and colleagues propose that the nongravitational acceleration is due to the gaseous expulsion of entrapped hydrogen from 'Oumuamua's water-rich icy body.

Which statement, if true, would most strongly support the claim made by Bergner and colleagues about the cause of 'Oumuamua's acceleration?

Answer

- A. Existing proposed models of outgassing from 'Oumuamua include the direct conversion of nitrogen or carbon monoxide from a solid to a gaseous state without becoming liquid, but these models have theoretical or observational inconsistencies.
- B. 'Oumuamua's trajectory is inconsistent with a nongravitational acceleration that would be caused by the release of hydrogen gas resulting from the processing of water ice (H_2O), but the interstellar object's observable properties can be explained if it has a significant component of molecular hydrogen ice (H_2).
- C. Since nongravitational accelerations of interstellar objects are several orders of magnitude weaker than gravitational accelerations, deviation from behavior that could be fully attributed to gravitational pull has been detected on a limited number of objects similar to 'Oumuamua.
- D. Exposure to interstellar cosmic radiation can result in the formation of embedded pockets of hydrogen gas in water ice; moreover, when traveling through the solar system, 'Oumuamua experienced warming sufficient to alter its icy structure and allow for outgassing.